

13th day of class - 10/7

Fun fact: If $E : y^2 = x^3 + Ax + B$, the j -invariant of E is defined to be

$$j(E) = \frac{6912A^3}{4A^3 + 27B^2}.$$

The denominator is zero only if E is singular.

If two elliptic curves $E : y^2 = x^3 + Ax + B$ and $E' : y^2 = x^3 + A_1x + B_1$ are isomorphic over $\mathbb{Q}$, then there is some nonzero rational number u so that $A_1 = Au^4$ and $B_1 = Bu^6$. This makes it so that $j(E) = j(E')$.

Conversely, if $j(E) = j(E')$, then E and E' are isomorphic over some extension of $\mathbb{Q}$ (in which A_1/A is a 4th power and B_1/B is a 6th power).

Last time: Torsion points in $E(\mathbb{C})$, the discriminant, and the statement of the Lutz-Nagell theorem.

Today: Section 2.4 - The proof of the Lutz-Nagell theorem.

Historical note: The theorem was discovered independently by Trygve Nagell and Élisabeth Lutz. Nagell published his proof in 1935, and Lutz published her proof in 1937 (in “Crelle’s journal”, the longest continuously published mathematics journal).

Definition. Let p be a prime number. If $r \in \mathbb{Q}$ and $r \neq 0$, write $r = \frac{a}{b}$ with $\gcd(a, b) = 1$. If p does not divide a or b , define $\text{ord}_p(r) = 0$. If $p|a$ and e is the highest power of p that divides a , define $\text{ord}_p(r) = e$. If $p|b$ and f is the highest power of p that divides b , define $\text{ord}_p(r) = -f$.

Context: Any nonzero rational number has a prime factorization (with positive and negative exponents). This $\text{ord}_p(r)$ is just the power of p that divides r . We call $\text{ord}_p(r)$ the p -adic valuation of r .

Strategy: Suppose that (x, y) is a point of finite order. Prove that $\text{ord}_p(x) \geq 0$ and $\text{ord}_p(y) \geq 0$ for all prime numbers p . This proves that no primes divide the denominators of x and y .

(When Andrew Wiles was working on something with elliptic curves, there was a certain real number $r \in \mathbb{R}$ expression he wanted to prove was zero. He showed $r = \Omega q$, where $\Omega \in \mathbb{R}$ and $q \in \mathbb{Q}$. Then he proved that there are infinitely many prime numbers that divide the numerator of q . This implies that $q = 0$.)

Step 1 - Think about valuations

Let $C : y^2 = x^3 + ax^2 + bx + c$ and $P = (x, y) \in E(\mathbb{Q})$. If $\text{ord}_p(x) = -\mu < 0$ and $\text{ord}_p(y) = -\sigma < 0$, then

$$\begin{aligned}\text{ord}_p(x^3 + ax^2 + bx + c) &= -3\mu \\ \text{ord}_p(y^2) &= -2\sigma.\end{aligned}$$

Since $3\mu = 2\sigma$, it follows that μ is even. Write $\mu = 2\nu$. Then we have $6\nu = 2\sigma$ and $\sigma = 3\nu$.

This shows that P is in $C(p^\nu) = \{(x, y) \in C(\mathbb{Q}) : \text{ord}_p(x) \leq -2\nu \text{ and } \text{ord}_p(y) \leq -3\nu\} \cup \{(0 : 1 : 0)\}$.

We have $C(p) \supseteq C(p^2) \supseteq C(p^3) \supseteq \dots$. The point at infinity is the only point in all of these sets. (This setup is often called a “filtration.”)

Claim 1: Each $C(p^\nu)$ is a subgroup of $C(\mathbb{Q})$!

Claim 2: If $P \in C(p^\nu)$ but P is not in $C(p^{\nu+1})$, then $mP \in C(p^{\nu+1})$ if and only if $p|m$. Moreover, $pP \in C(p^{\nu+1})$ but $pP \notin C(p^{\nu+2})$.

Example: Let $C : y^2 = x^3 - 12x + 20$. Then $C(\mathbb{Q}) \cong \mathbb{Z}$ and is generated by $P = (-2, 6)$.

We have

$$8P = (1/4, 33/8) \text{ and } 16P = (3001/1936, -192813/85184)$$

are both in $C(2)$. Thus,

$$24P = (62918249/2815684, 493493124949/4724717752)$$

is also in $C(2)$ since it is a subgroup.

Also, $8P \in C(2)$ but not in $C(4)$. This implies that $16P \in C(4)$ but not $C(8)$. Indeed, the denominators of the coordinates of $16P$ are $1936 = 2^4 \cdot 11^2$ and $85184 = 2^6 \cdot 11^3$.

Proof of Lutz-Nagell assuming claims:

We want to show that if $P \in E(\mathbb{Q})$ has finite order, then P is an integral point. This is true if and only if $P \notin C(p)$ for any prime p .

Suppose to the contrary that P has finite order and $P \in C(p)$.

Case I: The order m of P is coprime to p . Choose $\nu \geq 1$ so that $P \in C(p^\nu)$ but $P \notin C(p^{\nu+1})$. Then, by Fact 2 above, we have that $mP \in C(p^{\nu+1})$ if and only if $p|m$. However, $mP = (0 : 1 : 0) \in C(p^{\nu+1})$ but $p \nmid m$. This is a contradiction.

Case II: The order of m is a multiple of p . Let $P' = (m/p)P$ so that P' has order p . Choose ν so that $P' \in C(p^\nu)$ but $P' \notin C(p^{\nu+1})$. Then $pP' = (0 : 1 : 0)$, but the fact 2 above shows that pP' must actually be in $C(p^{\nu+1})$ but not in $C(p^{\nu+2})$. This is a contradiction because $pP' = (0 : 1 : 0) \in C(p^{\nu+2})$.

The t -map

If $P \in C(p)$, then there is some integer ν so that $P \in C(p^\nu)$ but not in $C(p^{\nu+1})$. Then

$$P = \left(\frac{a}{p^{2\nu}}, \frac{b}{p^{3\nu}} \right) = (x, y),$$

where $a, b \in \mathbb{Q}$ and p does not divide the number or denominator of a or b .

Define $t(P) = \frac{x}{y} = \frac{a}{p^{2\nu}} \cdot \frac{p^{3\nu}}{b} = p^\nu \frac{a}{b}$. We see that $\text{ord}_p(t(P)) = \nu$.

Also, since neither a nor b has p in the denominator, we can write $\frac{a}{b} = \frac{c}{d}$ where c and d are integers and $\gcd(c, p) = \gcd(d, p) = 1$. We can then reduce a/b modulo powers of p , because d has a multiplicative inverse mod p^k .

Theorem. *If $P_1, P_2 \in C(p^\nu)$, then*

$$t(P_1 + P_2) \equiv t(P_1) + t(P_2) \pmod{p^{3\nu}}.$$

Proof. I'm going to totally skip the details of this. (The details are on page 49-53 of the textbook.) Here's what goes into it. We let $t = x/y$ and $s = 1/y$. This is a change of variables that maps $(0 : 1 : 0)$ to $(0, 0)$. The elliptic curve becomes $C' : s = t^3 + at^2s + bts^2 + cs^3$. Lines get mapped to lines in this way.

Suppose we have $P_1 = (t_1, s_1)$ and $P_2 = (t_2, s_2)$ on C' and the line through them is $s = \alpha t + \beta$. Let $P_3 = P_1 * P_2 = (t_3, s_3)$. Then,

$$t_1 + t_2 + t_3 = -\frac{a\beta + 2b\alpha\beta + 3c\alpha^2\beta}{1 + a\alpha + b\alpha^2 + c\alpha^3}.$$

One can show that $\alpha \equiv 0 \pmod{p^{2\nu}}$ and $\beta \equiv 0 \pmod{p^{3\nu}}$. This shows that ord_p of the numerator is $\geq 3\nu$ and ord_p of the denominator is 0. $\square$

Example: Let $C : y^2 = x^3 + 17$. Then $C(\mathbb{Q}) \cong \mathbb{Z} \times \mathbb{Z}$ and is generated by $P_1 = (2, 5)$ and $P_2 = (-2, -3)$. We find that $3P_1 \in C(3)$ and $6P_2 \in C(3)$. This shows that $C(3) \supseteq \{3kP_1 + 6\ell P_2\}$. For $0 \leq x \leq 2$ and $0 \leq y \leq 5$ we find that $xP_1 + yP_2 \in C(3)$ if and only if $(x, y) = (2, 2)$ or $(x, y) = (1, 4)$. It follows from this that

$$C(3) = \{xP_1 + yP_2 : y \equiv 4x \pmod{6}\}.$$

Let $Q_1 = P_1 - 2P_2 = (106/9, 1097/27)$ and $Q_2 = 2P_1 + 2P_2 = (-8/9, -109/27)$. Now, $Q_1 + Q_2 = 3P_1 = (5023/3249, -842480/185193)$. Now

$$t(Q_1 + Q_2) = \frac{5023}{3249} \div \frac{-842480}{185193} = \frac{-286311}{842480} \equiv 3 \pmod{27}.$$

Also

$$\begin{aligned} t(Q_1) &= \frac{318}{1097} \equiv 6 \pmod{27} \\ t(Q_2) &= \frac{24}{109} \equiv 24 \pmod{27}. \end{aligned}$$

So indeed $t(Q_1 + Q_2) \equiv 3 \equiv 6 + 24t(Q_1) + t(Q_2) \pmod{27}$.

This looks like a homomorphism property. To make it a homomorphism, we have to do some stuff though.

Definition. Let $R = \{r \in \mathbb{Q} : \text{ord}_p(r) \geq 0\}$.

Note that R is closed under addition, because if

$$r_1, r_2 \in R$$

then $r_1 = \frac{a}{b}$ and $r_2 = \frac{c}{d}$ with $\gcd(a, b) = \gcd(c, d) = \gcd(b, p) = \gcd(d, p) = 1$. We get

$$r_1 + r_2 = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$

So the denominator is a divisor of bd , but b and d are both coprime to p . (The set R is also closed under multiplication.)

We don't have $t(P_1 + P_2) = t(P_1) + t(P_2)$ - this equality is only true modulo $p^{3\nu}$. Also, since P_1 and $P_2 \in C(p^\nu)$ we have $t(P_1) \equiv t(P_2) \equiv 0 \pmod{p^\nu}$.

So, what we do is we define $p^k R = \{p^k r : r \in R\}$. This is a subgroup of $p^k R$. So we can think of the map t as a homomorphism

$$t : C(p^\nu) \rightarrow p^\nu R / p^{3\nu} R.$$

The right hand side is a cyclic group of order $p^{2\nu}$.

Q: Is t surjective?

A: Not necessarily. There are examples where the map $t : C(p) \rightarrow pR/p^3R$ is just the zero map.

Q: What's the kernel of t ?

A: We have that P is in the kernel of $t : C(p^\nu) \rightarrow p^\nu R / p^{3\nu} R$ if and only if $t(P) \equiv 0 \pmod{p^{3\nu}}$. If we write $P = (a/p^{2k}, b/p^{3k})$ where $\text{ord}_p(a) = \text{ord}_p(b) = 0$, then $t(P) = \frac{a/p^{2k}}{b/p^{3k}} = p^k(a/b)$ so $\text{ord}_p(t(P)) = k$. Thus, $t(P) \equiv 0 \pmod{p^{3\nu}}$ if and only if $\text{ord}_p(t(P)) \geq 3\nu$, that is, if $P \in C(p^{3\nu})$.

Thus, $C(p^{3\nu})$ is the kernel of t .

So the first isomorphism theorem means that we have an injective homomorphism

$$C(p^\nu)/C(p^{3\nu}) \rightarrow p^\nu R / p^{3\nu} R.$$

Proof of claims

Claim 1: Each $C(p^\nu)$ is a subgroup of $C(\mathbb{Q})$.

Claim 2: If $P \in C(p^\nu)$ but P is not in $C(p^{\nu+1})$, then $mP \in C(p^{\nu+1})$ if and only if $p|m$. Moreover, $pP \in C(p^{\nu+1})$ but $pP \notin C(p^{\nu+2})$.

Proof of Claim 1. The congruence $t(P_1 + P_2) \equiv t(P_1) + t(P_2) \pmod{p^{3\nu}}$ only assumes that P_1 and P_2 are in $C(p^\nu)$. We have that $P \in C(p^\nu)$ if and only if $t(P) \equiv 0 \pmod{p^\nu}$. Thus, if P_1 and $P_2 \in C(p^\nu)$, then $P_1 + P_2 \in C(p^\nu)$. Also, if $P \in C(p^\nu)$ and $P = (x, y)$, then $-P = (x, -y) \in C(p^\nu)$ also. $\square$

Proof of Claim 2. If $P \in C(p^\nu)$ and $P \notin C(p^{\nu+1})$, then $\text{ord}_p(t(P)) = \nu$.

We have that $mP \in C(p^{\nu+1})$ if and only if $t(mP) \equiv 0 \pmod{p^{\nu+1}}$. But $t(mP) \equiv mt(P) \pmod{p^{3\nu}}$ and so $t(mP) \equiv mt(P) \pmod{p^{\nu+1}}$. This proves that $mP \in C(p^{\nu+1})$ if and only if $p|m$.

Now, we in fact have $t(mP) \equiv mt(P) \pmod{p^{\nu+2}}$ (since $3\nu \geq \nu + 2$ if $\nu \geq 1$). So if $t(P) \equiv 0 \pmod{p^\nu}$ but $t(P) \not\equiv 0 \pmod{p^{\nu+1}}$, then

$$t(pP) \equiv pt(P) \pmod{p^{\nu+2}}$$

must be $\equiv 0 \pmod{p^{\nu+1}}$, but $pt(P) \not\equiv 0 \pmod{p^{\nu+2}}$. Thus, $\text{ord}_p(t(pP)) = \nu + 1$ and so $pP \in C(p^{\nu+1})$ but $pP \notin C(p^{\nu+2})$. $\square$

Example (final exam from 2015): Let $C : y^2 = x^3 - 16x + 16$ and let $P = (0, 4)$.

(a) What is the smallest n so that $nP \in C(11)$?

We use Magma. The points kP for $1 \leq k \leq 6$ are integral. We keep checking and eventually find that $17P \in C(11)$ with $kP \notin C(11)$ for $1 \leq k \leq 16$. We can also compute that $t(17P) \equiv 110 \pmod{121}$.

(b) What is the smallest n so that $nP \in C(11^5)$?

We don't need Magma anymore.

We have that $17P \in C(11)$ but that $kP \notin C(11)$ for $1 \leq k \leq 16$. If $nP \in C(11)$, then if we write $n = 17q + r$ for $0 \leq r \leq 16$, then we have $nP = q(17P) + rP$. We know that $17P \in C(11)$ and thus $rP = nP - q(17P) \in C(11)$ since $C(11)$ is a group. This implies that $r = 0$. It follows that $C(11) = \{17qP : q \in \mathbb{Z}\}$.

Claim: For any positive integer k , we have that $nP \in C(11^k)$ if and only if $17 \cdot 11^{k-1} \mid n$.

We have proven the base case of $k = 1$.

Assume that we the k th case is known. Then $17 \cdot 11^{k-1}P \in C(11^k)$. Claim 2 implies that $17 \cdot 11^kP \in C(11^{k+1})$ and since $C(11^{k+1})$ is a subgroup of $C(\mathbb{Q})$, if $17 \cdot 11^k \mid n$, then $nP \in C(11^{k+1})$.

Conversely, if $nP \in C(11^{k+1})$, then $nP \in C(11^k)$ and so n is a multiple of $17 \cdot 11^{k-1}$. The inductive hypothesis implies that $17 \cdot 11^{k-1}rP \notin C(11^{k+1})$ unless $11 \mid r$ and from this it follows that $nP \in C(11^{k+1})$ if and only if $17 \cdot 11^k \mid n$.

QED Claim

It follows that the smallest n so that $nP \in C(11^5)$ is $n = 17 \cdot 11^4$.